

saved in s1 variable.)

The aim is to control the LED (for home appliance) with both touch and Wi-Fi. To do so, merge the functions

in Void loop(). An HTML page is created using HTML script in the code, as shown in Fig. 9.

You may change this code as per your application. You will see something like what is shown in Fig. 10 on the Web browser.

Power ESP32 with USB or 5V supply. Upload this code to ESP32. But, remember to select ESP32 DEV module and COM port from Tools menu before doing so.

Connections

There is one input (touchpad) and one output (LED1) in the circuit, as shown in Fig. 12. GPIO5 of ESP32 is connected to a resistor and from the resistor to the LED1. Touchpad is connected to GPIO4 of ESP32.

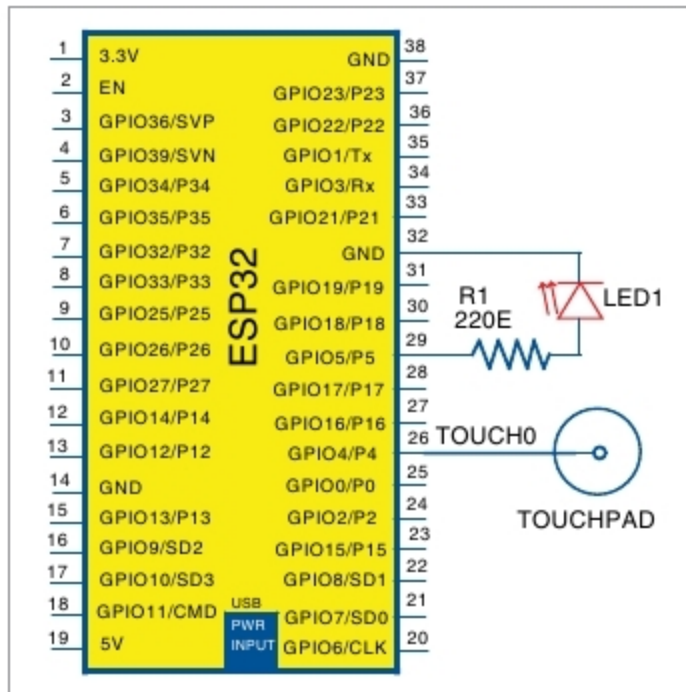


Fig. 12: Circuit connections with ESP-WROOM-32

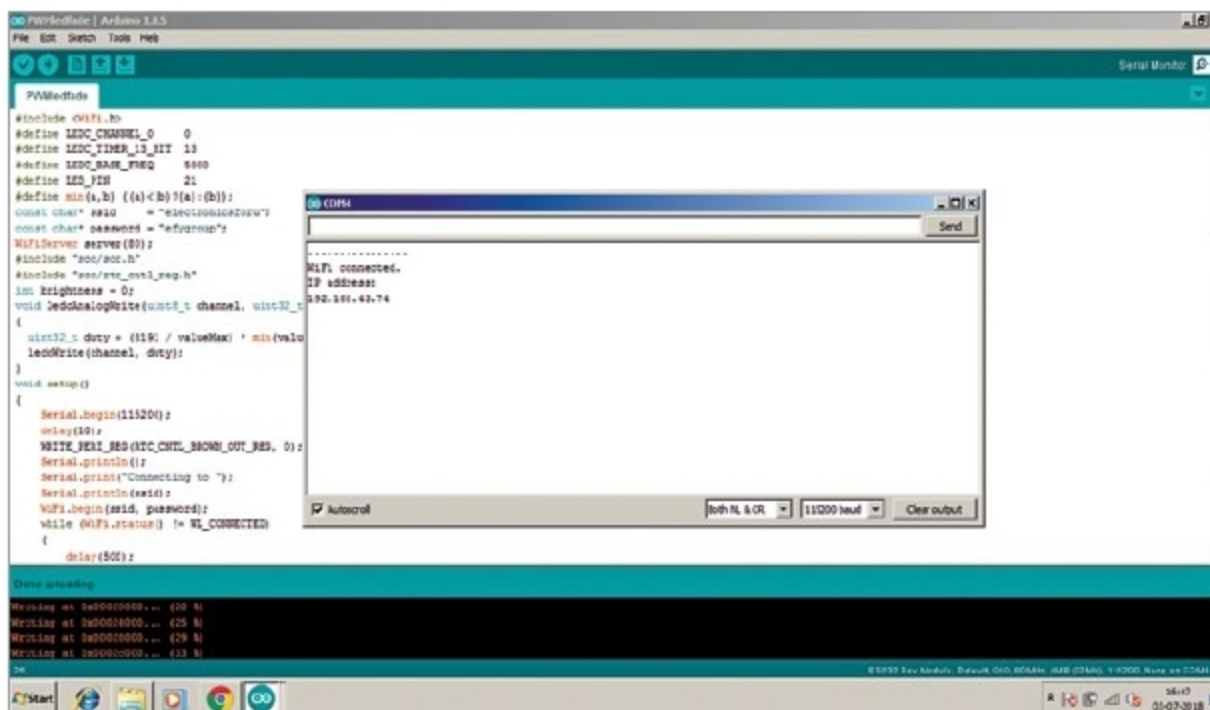


Fig. 13: Wi-Fi connected

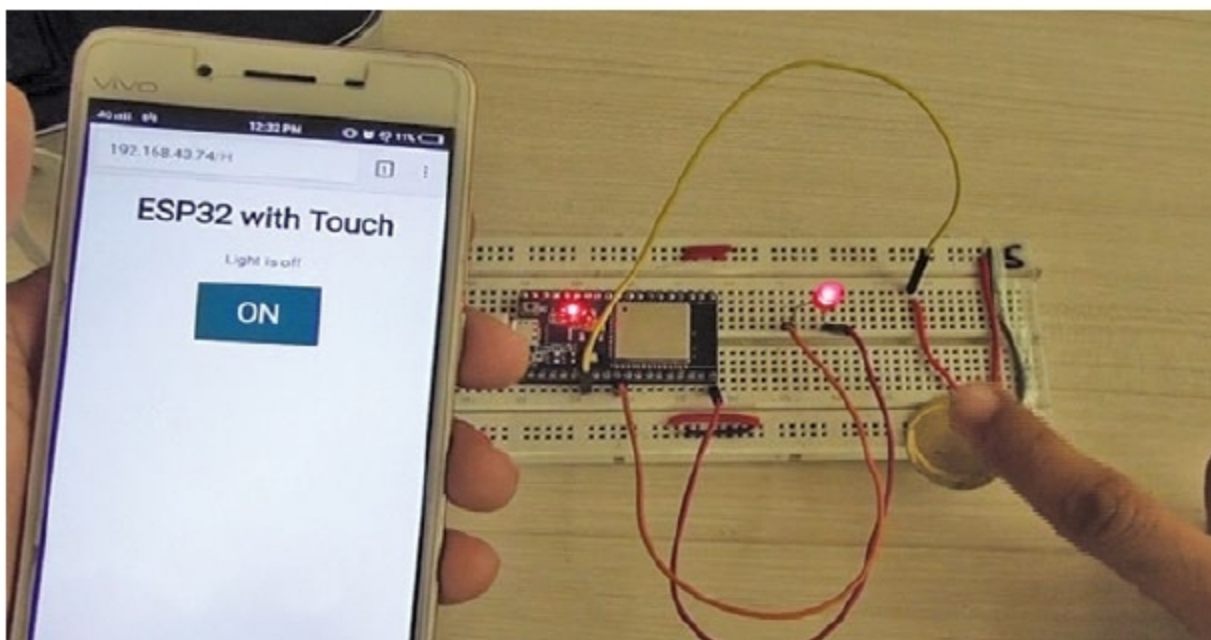


Fig. 14: Controlling LED using Wi-Fi and touch control

EFY Note

The source code of this project is available for free download at source.efymag.com



Fig. 15: (a) Install ESP32 board library and (b) Relay connections

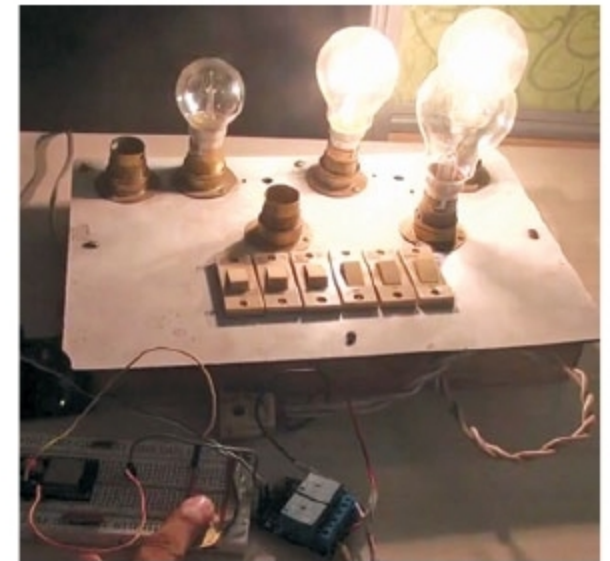


Fig. 16: Author's prototype

Connecting the Web server

After uploading the code, go to Tools > Serial Monitor in Arduino IDE. ESP32 will try to connect to Wi-Fi and display its IP address on Arduino serial monitor, as shown in Fig. 13.

Make sure that the Wi-Fi router to be connected is already open. Enter IP address (displayed in serial monitor) in the browser of the device (smartphone or laptop) connected to the same Wi-Fi, in the format given below.

<http://xxx.xxx.xx.xx> (your IP displayed in Arduino serial monitor)

HTML Web page, as designed in the main code, will open. Connect your smartphone to ESP32, and test Wi-Fi and touch input to control the LED1 (Fig. 14).

You may connect a relay instead of the LED1 to control the home appliance. For relay connections, scan the QR code given in Fig. 15(b).

The author's prototype is shown in Fig. 16. **EFY**

Kunal Verma is DIYer and community manager at EFY. His interests include the IoT, robotics and defence engineering